

Research Project (Physics)

Science

May, 2026

Research Project (Physics) (A14857)

Short Title: Research Project (Physics)
Faculty Science and Computing
Department Science
Cluster Placement and Projects
Author CWALSH
Credits 10

Level: Advanced

Description of Module / Aims

This module provides the student with the experience of working on an extended research project in collaboration with a project supervisor. The student will have selected their project and carried out a literature review as part of the 'Research Methodology and Literature Review' module in semester 7. The student will embark, under the guidance of his/her supervisor, on the development of the solution to a specified research problem. In addition to the students' supervisor, students will be timetabled for one full day in a dedicated project lab where lecturers will examine project-specific technical problems and provide support for the student. At the end of the module, the student will disseminate their findings through a dissertation, and will be expected to defend the project with a final presentation, including a demonstration where appropriate.

Programmes

RESA-0174 BSc (Hons) in Physics for Modern Technology (WD_KPHE_B)

4 / 8 / M

Indicative Content

- Planning and design of a scientific research project
- Implementation of research project
- Research project management
- Production of documentation associated with the project
- Presentation of practical aspects and outcomes of project

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Develop a disciplined approach to undertaking a research project.
2. Develop a range of skills in the application of physics, mathematics and computing to the solution of a research problem.
3. Plan a set of research activities; assess and design the appropriate equipment for the research task.
4. Select appropriate research methods, tools and techniques for the research task.
5. Appraise different research methodologies and be able to progress to postgraduate level study and research.
6. Produce a final dissertation which details the essential scientific and practical aspects and outcomes of the project to the appropriate deadline.
7. Produce a presentation and a demo where appropriate, which summarises the essential scientific and practical aspects and outcomes of the project.

Learning and Teaching Methods

- Consultations with project supervisor.
- Preparation of project dissertation.
- Oral presentation.

- Self-directed learning.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	30%	
<i>Supervisor's Report</i>	10%	1,2,3,4,5
<i>Presentation</i>	20%	7
Dissertation	70%	1,2,3,4,5,6,7

Assessment Criteria

- <40%:** Poor understanding of the scientific principles behind the research project, inadequate level of research work performed in terms of quality and quantity of research output. Poor level of communication with project supervisor. Poor quality of final report in terms of structure, referencing, readability, interpretation and analysis of results. Inadequate communication skills evident through a poor project presentation.
- 40%-49%:** Basic understanding of the scientific principles behind the research project, adequate level of research work performed in terms of quality and quantity of research output. Adequate level of communication with project supervisor. Acceptable quality of final report in terms of structure, referencing, readability, interpretation and analysis of results. Adequate communication skills evident through a project presentation.
- 50%-59%:** Good understanding of the scientific principles behind the research project, good level of research work performed in terms of quality and quantity of research output. Good level of communication with project supervisor. Good standard of final report in terms of structure, referencing, readability, interpretation and analysis of results. Good communication skills evident through a project presentation, supplemented by an ability to present the knowledge in a clear and precise manner.
- 60%-69%:** Very good understanding of the scientific principles behind the research project, very good level of research work performed in terms of quality and quantity of research output. Very good level of communication with project supervisor. Very good standard of final report in terms of structure, referencing, readability, interpretation and analysis of results. Very good communication skills evident by an ability to present the knowledge to a high technical standard
- 70%-100%:** All previous to an excellent level. Shows initiative and demonstrates an ability to abstract ideas and reflect on the process.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Practical	84	
Independent Learning	186	

Supplementary Material

- Alley , M. *_The Craft of Scientific Presentations: Critical Steps to Succeed and Critical Errors to Avoid_*. 2nd ed. USA: Springer, 2013.
- Alley, M. *_The Craft of Scientific Writing_*. 3rd ed. USA: Springer, 2003.

- Hofmann, A. _Scientific Writing and Communication: Papers__, Proposals, and Presentations_ . UK: Oxford University Press, 2013.

Requested Resources

- SCIENCE LAB: Physics